A BIRATIONAL MORPHISM IS AN ISOMORPHISM OVER A DENSE OPEN, UNDER QUASI-COMPACTNESS OR FINITENESS HYPOTHESES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A dense open over which a birational morphism satisfying these hypotheses becomes finite. Lemma \ref{lemma-finite-over-dense-open}, cited at proof lines 13383-13384. Reduces the assertion to a finite birational morphism.
- A birational morphism from an irreducible scheme onto an irreducible target has an open immersion as its restriction to a nonempty open. Lemma \ref{lemma-birational-birational}, cited at proof lines 13387-13388. Supplies the nonempty open of the source.
- The image of the complement of that open is closed because the morphism is finite. used at proof lines 13389-13391. Lets the closed subset be removed to get an isomorphism.

Source reference. Original lemma 0BAJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14212–14212. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0BAI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14211–14211. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BAC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14205–14205. Navigation pointer to a statement the

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Lemma 0.2. *Let X, Y be schemes. Let $f : X \to Y$ be a birational morphism between schemes which have finitely many irreducible components. Assume*

(1) *either f is quasi-compact or f is separated, and*
(2) *either f is locally of finite type and Y is reduced or f is locally of finite presentation.*

Then there exists a dense open $V \subset Y$ such that $f^{-1}(V) \to V$ is an isomorphism.

Proof. By Lemma 0BAI we may assume that f is finite. Since Y has finitely many irreducible components, we can find a dense open which is a disjoint union of its irreducible components. Thus we may assume Y is irreducible. By Lemma 0BAC we find a nonempty open $U \subset X$ such that $f|_U : U \to Y$ is an open immersion. After removing the closed (as f finite) subset $f(X \setminus U)$ from Y we see that f is an isomorphism. □